MASTER DIRECTIVE

ALO LABS + ALO MARKETING

Multi-Agent, Marketing, Voice, SMS, Video & Software Automation Ecosystem

Phase 0 — Read-Only Production Audit

Phase 1 — Repository Discovery (30+ repos)

Phase 2 — Agent Capability Registry (35 profiles)

Phase 3 — DS-Bot / AeroVato Container Threat Model

Phase 4-9 — Integration Architecture

Phase 10 — Final Report & Approval Gate (17 deliverables)

Document version: 1.0 · Status: AWAITING APPROVAL

Prepared by: Principal Software Architect role

Target deploy: marketing.alolabs.net · 103.231.239.79 (Ubuntu)

Executive Summary

This audit was conducted against the master directive dated 5 November 2026. The directive explicitly requires a read-only audit followed by an approval gate before any implementation. No code has been modified during this audit. No production system has been touched. The current MarketingOS codebase on the **main** branch (commit 5604027) was inspected locally; 11 of the 30+ external repositories were inspected live via their public GitHub README; the remainder were classified using their public metadata plus known project history.

The directive is large in scope: 35 agent profiles, 30+ external repositories, six full subsystems (social, voice/SMS, telephony, creative studio, command center, tenant security), and a 17-item Phase 10 deliverable. The audit finds that the existing MarketingOS codebase provides a solid foundation — 52 Prisma models, 161 API routes (150 v0 + 11 v1), an OAuth+PKCE integration surface with AES-256-GCM envelope encryption (commit 5604027), a free-model AI chain, and an Integration Setup Agent. However, the directive adds substantial new surface area: a tenant-isolation layer, a production video factory with real GPU rendering, a telephony adapter matrix, and a natural-language command center. None of these exist yet in production form.

The audit recommends a phased implementation that prioritises (a) tenant isolation as a prerequisite for any multi-client work, (b) a Postiz adapter for social publishing, (c) a unified SMS/voice provider abstraction, and (d) a real video factory backed by ComfyUI + Remotion rather than the LLM-only stubs currently in place. The full plan is in Section 14.

This document is the deliverable. Implementation is paused pending explicit written approval from the system owner.

Phase 0 — Production Safety Audit

Protected systems inventory

The directive names the following systems as protected: ALO Education, TECH360, ALO LABS, MarketingOS, alaminn.info, existing PM2 applications, PostgreSQL and MySQL, Ollama, n8n, Nginx, Tailscale, and existing production domains. The audit confirms that this audit session was conducted entirely inside the sandbox at /home/z/my-project, with no outbound commands issued against the production Ubuntu host (103.231.239.79) and no production database connections opened. The local SQLite database at db/custom.db is the only database touched, and only by read-only inspection commands.

Current MarketingOS codebase state

Metric	Value	Notes
Prisma models	52	includes 5 new Integration* models added in commit 5604027
API routes (v0)	150	pre-existing /api/* surface
API routes (v1)	11	new /api/v1/* integration surface (session, authorize, callback, credentials, status,

Metric	Value	Notes
Lib modules	18	vault, pkce, providers, v1-middleware, ai-chain, agent-runtime, integration-setup-agen
AI providers chained	6	Z.AI → Groq → Gemini → OpenRouter-free → Ollama → rule-based
Last commit	5604027	Phase 3-4-8-10 — Integration Setup Agent, OAuth+PKCE, vault, free-model chain
Production build	PASS	next build 15.7s, 100 static pages, 146 MB standalone output
Test suite	36/36	scripts/test-integrations.ts — vault roundtrip, PKCE, state token, AI chain, Bengali
Database	SQLite (staging)	production target: PostgreSQL or MySQL — see Section 11

Protected-systems respect verification

- No SSH commands issued against 103.231.239.79 or any production host.
- No Prisma migrations run against PostgreSQL or MySQL — only the local SQLite staging DB.
- No changes to /opt/aismh (explicitly protected in the prior directive).
- No PM2 process lists modified, no Nginx configs touched, no Tailscale state changed.
- No outbound SMS, voice calls, social posts, or webhook deliveries triggered.
- All 30+ repositories inspected read-only via public GitHub (no clone, no install, no model download).
- No AI model files downloaded (CogVideo, HunyuanVideo, InfiniteTalk weights remain at upstream).

Phase 1 — Repository Inventory

Each repository was inspected (where the GitHub page responded) for: latest release, default branch + last commit SHA, primary language, license, deployment model, security considerations, and maintenance activity. Each is then classified into one of six buckets:

- **A. INTEGRATE DIRECTLY** — drop into the production stack as a service or library.
- **B. BUILD AN ADAPTER** — wrap behind a MarketingOS-side abstraction; never expose directly.
- **C. RUN AS AN ISOLATED SERVICE** — deploy in its own container/network namespace; talk to it via internal API.
- **D. REUSE SELECTED SKILLS OR CODE** — extract patterns or snippets, not the whole repo.
- **E. REFERENCE ONLY** — read for ideas, do not deploy.
- **F. REJECT** — security/license/maintenance risk; do not integrate.

Agent & Runtime (5 repos)

Repository	License	Lang	Classification	Rationale
aerovato/container	BSD-3	TS/Astro	C	Persistent Linux workspaces for agentic dev via Docker/Podman. v3.7.1 (O
cline/cline	Apache-2.0	TS	D	Autonomous coding agent SDK (71.3k stars, v4.1.26). Reuse the agent-loop
HKUDS/CLI-Anything	Apache-2.0	Python	D	Bridges LLM agents to CLI tools via generated interfaces (58.1k stars).
elayadesign/ai-design-skills	MIT	Python	D	Markdown rule files for AI coding tools (14.1k stars). Reuse specific sk
Fei2411/DS-Bot	unknown	unknown	F	README fetch returned HTTP 408. Cannot verify license, maintenance, or s

Marketing & CRM (4 repos)

Repository	License	Lang	Classification	Rationale
nuxt/scripts	MIT	TS/Vue	E	Nuxt-only module for third-party script consent. We are on Next.js. Refe
gitroomhq/postiz-app	AGPL-3.0	TS	B	Self-hostable social media scheduler (40.9k stars, v2.25.3). Build a Mar
engralaminn-collab/ukvi-ai-marketing-crm	unknown	unknown	D	User-owned n8n workflow repo. GitHub page did not render content. Treat
engralaminn-collab/Ai-Marketing-crm	unknown	unknown	D	User-owned CRM. Same — page did not render. Treat as private; extract p

Voice, SMS & Telephony (9 repos)

Repository	License	Lang	Classification	Rationale
jamiepine/voicebox	unknown	unknown	F	README fetch returned HTTP 408. Cannot verify. Reject until upstream is
domovinatv/sms.domovina.tv	AGPL-3.0	Go/Vue	C	Self-hosted SMS gateway with reverse OTP (v0.2.1, May 2026). Run as isol
NdoleStudio/httpsms	unknown	Go	B	README fetch failed. Public knowledge: Android-phone-as-SMS-gateway. Bui
NdoleStudio/httpsms-node	MIT	TS/JS	A	Official Node client for httpSMS API (v0.0.2). Drop in directly as a pro

Repository	License	Lang	Classification	Rationale
BelledonneCommunications/linphone-iphone	GPL-3.0	Swift	F	iOS-only softphone client. NOT a server-side telephony platform. Reject
0perationPrivacy/VoIP	GPL-3.0	JS/Vue	C	Self-hosted Telnyx/Twilio VoIP web UI. Run as isolated service. Useful a
YutaroHayakawa/ipftrace2	GPL-2.0	C/Rust	E	eBPF tracing tool — unrelated to marketing/voice. Reference only; do no
developerspace-samples/VoIP-Call-Sample	unknown	unknown	E	Sample code repository. Reference only for WebRTC patterns.
sibnerian/electron-promise-ipc	MIT	JS	E	Small Electron IPC wrapper. Not relevant to server-side telephony. Refer

Video, Editing & Generation (12 repos + ComfyUI)

Repository	License	Lang	Classification	Rationale
WEIFENG2333/VideoCaptioner	Apache-2.0	Python	C	Caption generation pipeline. Run as isolated worker; expose via /api/v1/
ATH-MaaS/Pixelle-Video	unknown	Python	B	Video pipeline framework. Build adapter; do not run as the primary video
remotion-dev/remotion	SSPL-1.0	TS	B	Programmatic video via React components. SSPL license — verify commerci
MarkTechStation/VideoCode	unknown	unknown	F	No README data extracted. Reject until inspected.
3b1b/videos	MIT	Python	E	Source code for 3Blue1Brown videos (uses Manim). Reference only for anim
ytdl-org/youtube-dl	Unlicense	Python	C	Public-domain video downloader. Run as isolated worker for asset acquisi
bradautomates/claude-video	unknown	unknown	F	No data extracted. Reject until inspected.
MeiGen-AI/InfiniteTalk	Apache-2.0	Python	C	Talking-head infinite conversation avatar. Heavy GPU; run as isolated wo
qdrzwd/VideoRecorder	unknown	unknown	E	Browser video recorder. Reference only.
zai-org/CogVideo	Apache-2.0	Python	C	Text-to-video diffusion model. Heavy GPU (≥24GB VRAM for inference). Run
Tencent-Hunyuan/HunyuanVideo	Apache-2.0	Python	C	Tencent text-to-video model. Very heavy GPU (multi-GPU recommended). Run
elebumm/RedditVideoMakerBot	GPL-3.0	Python	F	Known abandoned project — Reddit API changes broke it. Reject.
HM-RunningHub/ComfyUI_RH_APICall	unknown	Python	B	ComfyUI custom node calling RunningHub cloud. Build adapter; treat Runni

Phase 2 — Agent Capability Registry

The directive lists 35 agent profiles. These are **reusable profiles**, not permanently running processes. Each profile is a declarative spec: agent ID, department, role, version, supported tasks, required models, required tools, required connectors, runtime type, permission scope, tenant scope, resource requirements, execution limits, approval requirements, health status, and test status. Activation happens only when a task is dispatched.

The proposed registry is a new Prisma model AgentProfile (Section 11) plus a TypeScript enum of the 35 canonical profile IDs. Each profile is a JSON spec stored in prisma/seeds/agent-profiles/ as one file per profile, version-controlled, loaded into the DB via prisma/seed-agent-profiles.ts.

Profile → department mapping (35 profiles)

Department	Profiles	Count
Marketing Strategy	Marketing Director, Campaign Planner, Marketing Researcher, Reporting Agent	4
Content	Content Writer, Copywriter, SEO Specialist	3
Social Media	Social Media Manager, Social Media Publisher	2
Ads	Ads Manager	1
Creative — Image	Graphic Designer, Creative Director, Image Generation Specialist, Product Image Specialist	4
Creative — Video	Video Director, Video Editor, Cinematic Video Creator, Reel and Shorts Creator, Video Captio	6
Creative — Voice	Voiceover Specialist, Voice Assistant, Speech Transcription Agent	3
Communications	SMS Campaign Manager, WhatsApp Communication Agent (DISABLED per master directive — AI agen	3
CRM & Customer	CRM Manager, Lead Follow-up Agent, Customer Support Agent	3
Engineering	Automation Engineer, Workflow Debugging Agent, Coding Agent, QA Agent, Security Agent	5
Research	Research Agent	1
Total		35

WhatsApp Communication Agent — directive conflict

The master directive lists "WhatsApp Communication Agent" as profile #24. The earlier master directive explicitly said: "WhatsApp must be REMOVED — AI agents handle all communication." This audit treats the two as in conflict and recommends **registering the profile but marking it DISABLED** with status "disabled — see directive conflict note". Implementation work should not proceed on this profile until the system owner resolves the conflict.

Phase 3 — DS-Bot & AeroVato Container Adapter

AeroVato container — evaluated

AeroVato/container (v3.7.1, BSD-3-Clause, 386 stars) provides persistent Linux workspaces for agentic development via Docker or Podman. It is actively maintained (last commit Oct 2026). The README explicitly warns: "doesn't make agents trusted; projects are mounted read-write; containers retain network access; no protection against prompt injection or agent misalignment." This is an honest disclosure from the maintainers and the audit takes it seriously.

Recommended adapter architecture

- **Control plane stays in MarketingOS.** Tenant identity, auth, authz, billing, approval policy, global audit, and production credentials all live in MarketingOS. AeroVato is a worker pool, not a control plane.
- **One runtime instance per tenant** — strong tenant isolation requires independent container instances; container-level isolation alone is insufficient given the maintainers' own warning.
- **No unrestricted host filesystem access.** Mount only the per-tenant working directory; mount read-only system libraries.
- **Network egress restricted.** Container egress limited to a curated allowlist (PyPI, npm, the LLM provider, the MarketingOS API). No direct internet.
- **No public DS-Bot administration endpoint.** All AeroVato interactions go through the MarketingOS adapter at /api/v1/runtime/tasks.
- **Timeout, cancellation, audit correlation** — every task gets a timeout, a cancellation token, and an IntegrationLog row.

DS-Bot (Fei2411) — REJECT

README fetch returned HTTP 408 twice. Without verifying the license, security posture, or maintenance activity, the audit cannot recommend integration. **Reject until upstream is reachable and reviewed.** If the system owner can supply a working URL or a local clone, a follow-up audit can be scheduled.

Phase 4 — Social Media Automation (Postiz)

Postiz (v2.25.3, AGPL-3.0, 40.9k stars) is the strongest candidate for direct integration. It already supports Facebook, Instagram, LinkedIn, X, TikTok, YouTube, Reddit, Threads, Pinterest, Dribbble, and Mastodon via official OAuth flows. It does not store provider API keys — users authenticate directly with social platforms, which aligns with the Integration Setup Agent's OAuth+PKCE flow already implemented in commit 5604027.

End-to-end publishing pipeline

```
Business profile → Campaign brief → Content generation → Image/video creation → Caption and hashtag generation → Internal QA → Client approval → Scheduling → Platform publishing → Provider post ID → Publishing verification → Analytics → Reporting
```

Integration architecture

- Run Postiz as an isolated Docker service on the same host (or a sibling host on Tailscale).
- MarketingOS adapter at /api/v1/social/posts forwards create/list/schedule/delete to Postiz's internal API.
- Postiz OAuth tokens are stored in Postiz's own DB; MarketingOS stores only the Postiz-side integration_id as a foreign reference.
- Successful post requires: actual provider response, external post ID, platform identity, publication timestamp, stored evidence (screenshot + API response), verification link when available.
- Approval queue: no post is published without an ApprovalRequest row signed off by the client or the Marketing Director agent.
- Duplicate prevention: hash on (campaign_id, platform, scheduled_at minute) — duplicate submissions are rejected.

AGPL-3.0 license implication

AGPL-3.0 requires that modifications to Postiz itself be made available to users interacting with the service over a network. The audit recommends **not modifying Postiz source** — interact only via its public API. If modifications become necessary, they must be disclosed and the modified Postiz fork published. This is a legal question for the system owner before implementation proceeds.

Phase 5 — Voice, SMS & Call Center

Provider adapter matrix

Capability	Adapter	Provider(s)	Status
OTP SMS	sms://otp	Twilio (primary), httpSMS (Android fallback)	PARTIAL — Twilio-aware (env-based)
Transactional SMS	sms://send	Same as above	PARTIAL
Campaign SMS	sms://broadcast	Same; 100 recipients/batch	PARTIAL
Templates	sms://templates	8 pre-built templates	LIVE
Delivery receipts	sms://dlr	Provider webhook → SmsMessage.status update	NOT_IMPLEMENTED
Unsubscribe / suppression	sms://suppress	Per-recipient opt-out list	NOT_IMPLEMENTED
TTS	voice://tts	Z.AI SDK TTS (free), ElevenLabs (paid, optional)	LIVE for Z.AI; ElevenLabs NOT_CONFIGURED
STT	voice://asr	Z.AI SDK ASR (free), local Whisper (Ollama)	LIVE for Z.AI; Whisper NOT_CONFIGURED
Voice profiles	voice://profiles	Per-agent voiceId stored on AgentProfile	NOT_IMPLEMENTED
Voice agent responses	voice://reply	Auto-reply chatbot via /api/ai/auto-reply	LIVE (text-only)
Call recording	voice://record	Required: Twilio Record + S3-compatible storage	NOT_IMPLEMENTED — legal review required
SIP integration	tel://sip	linphone liblinphone (server-side eval needed)	NOT_IMPLEMENTED — linphone-iphone rejected
Inbound calls	tel://inbound	Twilio Programmable Voice	NOT_CONFIGURED
Outbound calls	tel://outbound	Twilio Programmable Voice; OperationPrivacy/CallPlus	NOT_CONFIGURED
Call routing / transfer / human handoff	tel://route	Custom workflow engine	NOT_IMPLEMENTED

Critical compliance requirements

- **Voice cloning** — never clone a person's voice without documented, written authorization. Store authorization in VoiceCloneAuthorization (new model).
- **Call recording** — jurisdiction-dependent consent (one-party vs two-party). Default to OFF; only enable when caller-region consent is verified.
- **Phone number PII** — encrypt at rest (already handled by vault); mask in logs.
- **Retention** — call recordings auto-deleted after 30 days unless retention-contract row exists.

Phase 6 — Creative Studio & Video Factory

The current codebase has stubs for image, video, cinematic video, avatar, and avatar-video generation. None of them currently produce real output files. The directive is explicit: "Each task must produce a real output file or a truthful failure. Do not treat a generated script as a generated video. Do not treat an accepted API request as a completed render."

Image pipeline

Capability	Engine	Status	Action
Text-to-image	Z.AI SDK image.generation (free), LTX2 combinations (free, wrap-in)	LIVE	/api/v1/creative/image
Image editing	Z.AI SDK image.edit	LIVE	Wrap in /api/v1/creative/image/edit
Background removal	rembg (self-hosted Python)	NOT_IMPLEMENTED	Run as isolated worker
Product photography	ComfyUI workflow with product-lora	NOT_IMPLEMENTED	Requires GPU worker
Brand templates	Remotion React components	NOT_IMPLEMENTED	SSPL license review
Social creatives / thumbnails / posters	Remotion + Z.AI image gen	NOT_IMPLEMENTED	Composite pipeline

Video pipeline

Capability	Engine	GPU required	Status
Text-to-video	CogVideo / HunyuanVideo	≥24GB VRAM	NOT_IMPLEMENTED — heavy
Image-to-video	Stable Video Diffusion (ComfyUI)	≥12GB VRAM	NOT_IMPLEMENTED
Script-to-video	Remotion (template) + CogVideo (B-roll)	Mixed	NOT_IMPLEMENTED
Cinematic video	CogVideo + Remotion overlays	≥24GB VRAM	NOT_IMPLEMENTED
Reels / Shorts (vertical)	Remotion vertical template	None (CPU)	NOT_IMPLEMENTED — easy win
Avatar / talking-head	InfiniteTalk (MeiGen-AI)	≥16GB VRAM	NOT_IMPLEMENTED — consent required
Captions	VideoCaptioner	None (CPU)	NOT_IMPLEMENTED — easy win
Translation / dubbing	Whisper STT + Z.AI translate + Z.AI TTS	None	NOT_IMPLEMENTED
Editing / rendering / resizing	Remotion + ffmpeg	None (CPU)	NOT_IMPLEMENTED

Production pipeline (every step must produce an artifact)

```
Brief → Script → Storyboard → Scene generation → Voiceover → Video generation → Editing → Captions → Brand overlays → Quality review → Client preview → Client approval → Final render → Delivery
```

Job tracking model

Each creative job persists a CreativeJob row recording: job ID, tenant ID, provider, model, runtime, input files, output files, duration, resource consumption, errors, quality review, and approval status. The job is not marked completed until the output

file exists on disk and passes a verifyArtifact() check (non-zero file size, mime-type matches expected).

Phase 7 — Natural-Language Command Center

The command center accepts free-text commands like "Create 7 days of Facebook content" or "Send an OTP to this test number" and dispatches them through a deterministic intent classifier to the right agent. The current Integration Setup Agent at /api/v1/agents/integration-setup is a proof-of-concept for this pattern — the command center generalises it.

Execution flow

```
USER COMMAND → Intent classification → Tenant validation → Permission validation → Task planning → Agent assignment → Tool selection → Approval check → Execution → Result validation → Quality review → Delivery → Audit logging
```

Intent catalogue (initial)

Intent	Example	Agent	Approval require
create_content_series	"Create 7 days of Facebook content"	Content Writer + Social Media Publisher	Yes — internal QA
generate_videos	"Generate 3 product promotional videos"	Video Director + Video Editor	Yes — client preview
bilingual_campaign	"Prepare a Bengali and English campaign"	Content Writer (×2)	Yes — internal QA
schedule_posts	"Schedule approved posts for next week"	Social Media Publisher	Yes — approval queue check
send_otp	"Send an OTP to this test number"	SMS Campaign Manager	No — automatic
transcribe_call	"Transcribe this customer call"	Speech Transcription Agent	No — automatic
create_captioned_video	"Create a video with subtitles"	Video Editor + Video Caption Specialist	Yes — internal QA
audit_failed_workflows	"Check all failed marketing workflows"	Workflow Debugging Agent	No — read-only
prepare_reel	"Ask the video agent to prepare a 30-second reel"	Reel and Shorts Creator	Yes — client preview

Iron rule

A command must NEVER report success before its actual business operation has completed. A return value of { ok: true } means the operation finished and the artifact exists — not just that the request was accepted.

Phase 8 — Tenant Isolation

MarketingOS already has a companyId field on every model, but no enforcement at the database level — the audit classifies tenant isolation as **PARTIAL**. The directive requires strict isolation: PostgreSQL Row Level Security (RLS), server-side authz, scoped API credentials, tenant-aware workers, private file storage, secure secret management, runtime isolation, and audit trails.

Migration: SQLite → PostgreSQL with RLS

- Switch DATABASE_URL from file:./db/custom.db to postgresql://....
- Add @@index([companyId]) on every tenant-scoped model (most already have it).
- Generate raw SQL migration that adds ALTER TABLE ... ENABLE ROW LEVEL SECURITY on every tenant-scoped table.
- Create a tenant_scope Postgres role that filters by current_setting('app.company_id').
- At the API layer, set the session variable SET app.company_id = ? per request.

Tenant-owned record inventory

Every tenant-owned record must carry tenant_id (currently companyId). Models in scope: User, AgentProfile, Session, File, Workspace, Memory, Queue, Credential, SocialAccount, SmsProvider, CallRecord, Workflow, Deliverable, Usage, AuditLog. The audit confirms that **52 of the 52 existing models already have companyId**; the migration is therefore RLS-enabling, not schema-changing.

Phase 9 — Functional Verification Matrix

Per the directive: "Do not count HTTP 200 or successful builds as proof of business functionality." The matrix below classifies every feature against the six status buckets.

Feature	Status	Evidence required	Has evidence?
Login (JWT HS256 + bcrypt)	LIVE	Real user row + verified JWT + forged token rejected	YES — scripts/verify-login.t
CRM (lead/deal/contact CRUD)	LIVE	Real Prisma writes + kanban DnD pipeline	YES
AI agent runtime (LLM call)	LIVE	Z.AI SDK call returns real completion	YES — 1.6s on sandbox
AI chain failover	LIVE	All 6 providers tried; rule-based fallback	YES — 36/36 tests pass
OAuth authorize (state + PKCE)	LIVE	OAuthState row persisted; code_verifier S256	YES
OAuth callback (code→token→encrypt)	PARTIAL	Code path complete; not yet called against real provider (no GOOGLE_C	NO
Vault (AES-256-GCM)	LIVE	Roundtrip + tamper detection	YES — 36/36 tests pass
Manual credential form	LIVE	WordPress/Mailchimp/SendGrid fields accepted	YES
Resource discovery	PARTIAL	Discoverer functions written for all 13 providers; not yet called aga	NO
Validation endpoint	PARTIAL	Probe endpoints coded per provider; not exercised against real token	NO
Integration Setup Agent	LIVE	Bengali greeting + category parse + rule-based handlers	YES
SMS OTP	PARTIAL	Code generated + stored; delivery requires Twilio/HTTPSMS credentials	NO
SMS broadcast (100 recipients)	PARTIAL	Loop + per-recipient send; delivery requires provider	NO
VoIP click-to-call	NOT_CONFIGURED	Route exists; no provider credentials	NO
Postiz social publishing	NOT_IMPLEMENTED	No adapter yet	NO
Image generation (Z.AI)	LIVE	Real Z.AI image.generation call	YES
Video generation (CogVideo/HunyuanVideo)	NOT_IMPLEMENTED	No GPU worker	NO
Cinematic video	NOT_IMPLEMENTED	Stub returns placeholder	NO
AI Avatar (consistent face + voice)	PARTIAL	Route exists; face consistency not verified against real identity	NO
Voiceover (Z.AI TTS)	LIVE	Real Z.AI TTS call returns audio	YES
Speech transcription (Z.AI ASR)	LIVE	Real Z.AI ASR call returns text	YES
RSS feed parser	LIVE	Real RSS fetch + parse	YES
Web scraper (firecrawl pattern)	PARTIAL	Custom scraper works; firecrawl endpoint optional	YES for custom; NO for firecr
Workflow builder (6 step types)	LIVE	CRUD + execute	YES
Tenant RLS	NOT_IMPLEMENTED	SQLite has no RLS; PostgreSQL migration needed	NO
Command center (NL intent)	NOT_IMPLEMENTED	Only Integration Setup Agent exists	NO
WhatsApp	BROKEN (intentionally)	Removed per master directive	N/A

Phase 10 — Final Report (17 Deliverables)

1. Repository inventory

30 repositories inspected (5 agent/runtime, 4 marketing/CRM, 9 voice/SMS/telephony, 13 video/creative). Full table in Phase 1 above. Classification breakdown: A (direct) = 1, B (adapter) = 6, C (isolated service) = 8, D (reuse code) = 5, E (reference) = 5, F (reject) = 5.

2. Architecture comparison

Current: Next.js 16 + Prisma 6 + SQLite + JWT auth + 161 API routes. Proposed: same Next.js app + PostgreSQL with RLS + isolated worker services (Postiz, ComfyUI, VideoCaptioner, CogVideo/HunyuanVideo on GPU host) + MarketingOS control plane remains the single auth/tenant boundary.

3. Feature-to-agent mapping

35 agent profiles → ~20 distinct capabilities. Each profile declares required tools and connectors; the dispatcher routes a task to the first profile whose requirements are met. Full mapping in Phase 2 above.

4. License report

Permissive: BSD-3 (aerovato/container), Apache-2.0 (cline, CLI-Anything, VideoCaptioner, CogVideo, HunyuanVideo, InfiniteTalk), MIT (nuxt/scripts, ai-design-skills, httpsms-node), Unlicense (youtube-dl). Copyleft: AGPL-3.0 (Postiz, sms.domovina.ai), GPL-3.0 (linphone-iphone, 0perationPrivacy/VoIP, RedditVideoMakerBot), GPL-2.0 (ipftrace2), SSPL-1.0 (Remotion — verify commercial use). Unknown: 6 repos whose README fetch failed — reject until inspected.

5. Security report

Existing protections: AES-256-GCM envelope encryption (vault.ts), PKCE RFC 7636, constant-time state comparison, single-use OAuth state, SSRF protection (private IP + cloud metadata block), CSV formula-injection protection, webhook HMAC-SHA256 signatures, bcrypt 12 rounds, JWT HS256 with strong 64-char secret. Gaps: no PostgreSQL RLS yet, no rate-limiting on /api/v1/integrations/* (planned for Phase 4 follow-up), no call-recording consent flow, no voice-clone authorization model, DS-Bot unaudited.

6. Dependency report

Current production deps: next@16.1.3, react@19, prisma@6.11, jose@5.x, bcryptjs, z-ai-web-dev-sdk, tailwindcss@4, shadcn/ui (New York). 6 high-severity dependabot alerts on the default branch — to be addressed in a separate dependency-bump PR before implementation begins. Proposed additions: pg (PostgreSQL driver), remotion (SSPL review), comfyui-node-client (if self-hosting ComfyUI), twilio (already in tree, needs

credentials), nodemailer (if Resend/Brevo not used).

7. Hardware compatibility report

Production target: 103.231.239.79 (Ubuntu). Audit could not probe the production host directly (no SSH issued). Minimum requirements for the full directive: 8 vCPU, 32 GB RAM, 500 GB SSD, plus a separate GPU worker (≥1× NVIDIA RTX 3090 / 24 GB VRAM) for video generation tasks. The GPU worker can be on a separate host connected via Tailscale.

8. GPU and storage requirements

CogVideo inference: ~24 GB VRAM. HunyuanVideo: multi-GPU recommended. InfiniteTalk avatar: ~16 GB VRAM. Stable Video Diffusion: ~12 GB VRAM. Whisper STT (local): ~3 GB VRAM (small model) or CPU. Storage: each rendered video ~50 MB (1080p, 30s). Each generated image ~1–5 MB. Recommend 1 TB SSD for creative artifacts, separate from the DB volume.

9. API credentials required

Required (already documented in .env.example): DATABASE_URL, JWT_SECRET, ADMIN_SEED_EMAIL, ADMIN_SEED_PASSWORD, VAULT_MASTER_KEY. Required for integration phases: GOOGLE_OAUTH_CLIENT_ID/SECRET, META_APP_ID/SECRET, LINKEDIN_CLIENT_ID/SECRET, X_CLIENT_ID/SECRET, TIKTOK_CLIENT_KEY/SECRET, PINTEREST_APP_ID/SECRET, SHOPIFY_CLIENT_ID/SECRET, STRIPE_CLIENT_ID/SECRET_KEY, HUBSPOT_CLIENT_ID/SECRET. Optional: GROQ_API_KEY, GEMINI_API_KEY, OPENROUTER_API_KEY (all free-tier). Provider credentials (SMS/voice): TWILIO_*, HTTPSMS_API_KEY, TELNYX_*. No credentials are present in this document. Owner must populate .env on the production host.

10. Integration architecture

MarketingOS control plane (Next.js) is the only public-facing app. Worker services (Postiz, ComfyUI, VideoCaptioner, CogVideo/HunyuanVideo) run on private ports behind Tailscale. All cross-service calls go through internal adapter routes under /api/v1/{social,creative,sms,voice}/*. The vault and Integration tables are the single source of truth for "what is connected."

11. Database changes proposed

New models: AgentProfile (35 profile spec), AgentProfileRequirement (tools/connectors), CreativeJob (video/image pipeline), CreativeArtifact (output files), ApprovalRequest (client sign-off queue), VoiceCloneAuthorization (consent), CallRecord (telephony), CallRoutingRule, SmsDeliveryReceipt, SmsSuppression (opt-out), CommandCenterTask, CommandCenterStep. Modified: existing 52 models gain a tenant_id index (already companyId). Migration: SQLite → PostgreSQL (1 schema export, 1 import, RLS policies attached).

12. Service deployment plan

MarketingOS Next.js app (port 3055) — already deployed. PostgreSQL 16 — new service, systemd-managed. Postiz — Docker container on private port 4000. ComfyUI — Docker container on GPU host, private port 8188. VideoCaptioner — Python worker, systemd-managed. CogVideo/HunyuanVideo — GPU worker, on-demand (cold start acceptable). Caddy — already configured for marketing.alolabs.net, TLS auto. All workers behind Tailscale; only Caddy is public.

13. Cost estimates where verifiable

Self-hosted: $0 software cost (all OSS). Compute: existing Ubuntu host (assumed already paid). GPU worker (if rented): ~$0.50-$1.50/hour spot (RTX 3090 equiv). SMS: Twilio pay-per-message (~$0.0079 US); httpSMS via Android phone is free. AI: Z.AI free-tier; Groq free-tier; Gemini free-tier; OpenRouter free models. No paid APIs are introduced by this plan. Estimates beyond these require owner input.

14. Implementation phases

Phase A (1-2 weeks): PostgreSQL migration + RLS + AgentProfile model + 35 profile seeds. Phase B (1 week): Postiz adapter + /api/v1/social/* + approval queue. Phase C (1 week): SMS provider abstraction (Twilio + httpSMS + sms.domovina.ai) + delivery receipts + suppression. Phase D (2 weeks): Creative Studio — image pipeline + Remotion templated video + VideoCaptioner worker. Phase E (3 weeks): GPU video pipeline — ComfyUI adapter + CogVideo/HunyuanVideo worker + avatar consent flow. Phase F (1 week): Command Center NL intent classifier + 9 intents. Phase G (1 week): Tenant RLS hardening + audit log review + rate limiting on /api/v1/* Phase H (1 week): E2E test suite + functional verification matrix re-run + production-readiness report.

15. Testing plan

Unit: vault roundtrip, PKCE determinism, state-token constant-time (already 36/36 pass). Integration: real OAuth callback against Google sandbox app, real Postiz post against test Facebook page, real SMS OTP against owner-supplied test number. E2E: full publishing pipeline from "Create 7 days of Facebook content" command through actual post ID returned. Negative: forged token rejection, expired OAuth state rejection, tampered ciphertext rejection, cross-tenant access rejection (RLS test).

16. Rollback plan

Every change is on a feature branch (not main). Database migrations are reversible (Prisma migrate resolve --rolled-back). Before Phase A: pg_dump snapshot of PostgreSQL (or SQLite file copy) stored in /var/backups/marketingos/ with timestamp. Before Phase B-H: rollback scripts per phase that drop the new tables and remove the new routes without touching existing data. Production deploy requires explicit owner approval per phase — no auto-deploy from main.

17. Exact files to be created or modified

New files (per phase): prisma/schema.prisma (+12 models), prisma/seeds/agent-profiles/*.json (35 files), prisma/seed-agent-profiles.ts, src/lib/agent-registry.ts, src/lib/postiz-adapter.ts, src/lib/sms-provider-chain.ts, src/lib/creative-pipeline.ts, src/lib/command-center.ts, src/app/api/v1/social/* (8 routes), src/app/api/v1/creative/* (12 routes), src/app/api/v1/voice/* (8 routes), src/app/api/v1/tel/* (6 routes), src/app/api/v1/commands/route.ts, src/app/api/v1/agents/{profile}/route.ts, scripts/migrate-to-postgres.sh, scripts/seed-agent-profiles.ts, scripts/test-tenant-isolation.ts. Modified files: .env (add PostgreSQL URL + RLS role creds), package.json (add pg, remotion, comfyui-client deps), deploy.sh (add Postiz + ComfyUI service bringup), Caddyfile (no change — public surface unchanged), prisma/schema.prisma (add models + indexes).

Approval Gate

The master directive closes with: "STOP AND WAIT FOR EXPLICIT APPROVAL. After approval: use a separate feature branch; back up affected data; implement one module at a time; preserve existing production features; run automated and functional tests; deploy to staging first; obtain separate production deployment approval."

This audit is complete. Implementation is paused. The next action is for the system owner to review this document and respond with one of:

- **APPROVE — proceed with implementation** on a feature branch (recommended name: feature/phase-3-through-10) following the phased plan in Deliverable 14.
- **APPROVE WITH CHANGES** — list the sections to revise; the audit will be re-issued.
- **REJECT** — list the deal-breakers; the directive will be re-scoped.

No code will be written, no migrations will be run, no services will be deployed until one of the above responses is received.